

00 — Executive Summary

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— consolidated from SPECIFICATION.md and MASTER_INDEX.md.

One-sentence pitch

HelixVPN is a **self-hostable overlay network with a privacy-VPN front end**: Cloudflare Tunnel + WARP, rebuilt as Tailscale-style coordination, with Mullvad's obfuscation stack, fully self-hostable, on one shared codebase.

Why it exists

Remote users need full, policy-scoped access to **one or many** internal/home/lab networks **without any inbound port-forward**. Internal hosts dial *outbound* to a public gateway; the gateway relays and routes.

Lead differentiators

1. **Self-hosted + Mullvad-grade obfuscation** (MASQUE/QUIC, Shadowsocks, UDP-over-TCP, LWO, DAITA).
2. **One user → N joined private networks** with bidirectional gateway and ACL routing.
3. **8-platform reach** including Aurora OS and HarmonyOS NEXT on one codebase.
4. **One shared Rust + Flutter codebase** — client, connector, and edge reuse the same core.
5. **Event-driven real-time control plane** — sub-second convergence, push-not-poll.

The three roles

Role	Runs	Direction	Reuses
Connector	Inside a private network	Outbound-only	helix-core advertise/route mode
Gateway	Public VPS	Accepts dials	helix-core edge + helix-go control plane
Client	End-user device	Outbound-only	helix-core + helix-ui

The three app classes

App	User	Platforms
Helix Access	End user	iOS, Android, Aurora, HarmonyOS, Windows, Linux, macOS, Web (limited)
Helix Connector	Network operator	Linux/Windows/macOS daemon (+ optional UI); Android/embedded appliance
Helix Console	Admin	Web + Desktop Flutter (API client only, no tunnel core)

Non-negotiable principles

1. Control plane and data plane are strictly separated — **fail-static**.
2. **WireGuard** is the cryptographic core; transports are pluggable *beneath* it.
3. **Outbound-only** from edges — no inbound holes.
4. **Push, don't poll** — WatchNetworkMap streaming deltas.
5. **One core per concern, reused everywhere**.
6. **Self-hostable by one person, scalable to many gateways**.
7. **No-logging by construction** — CI schema-lint fails on durable flow tables.
8. **Schema-first, zero drift** — protobuf + OpenAPI, generated clients.
9. **Decoupled, reusable components** per constitution §11.4.28/§11.4.74.

Phase roadmap

Phase	Goal	Hard gates
Phase 0 — Spike (~3-4 weeks)	Prove the hard parts on production interfaces	G1-G6 (throughput, MASQUE, iOS memory, Go-vs-Rust edge, FFI, reconcile)
Phase 1 — MVP	Self-hostable product	8-criteria Definition-of-Done (self-host, enroll, reach/deny, MASQUE, policy <1s, revoke <1s, kill-switch, no logs)
Phase 2 — Parity + Reach	Full Mullvad parity + HA/multi-region	P2-AC1-AC8 + P2-SLO1-SLO5 (P2P, DAITA, multi-hop, PQ, desktop, GitOps, regional failover <3s)
Phase 3 — Extended Reach	HarmonyOS, Aurora, WASM proxy, billing, audit, reproducible builds	G20-G26

Open decisions (D1-D8)

See [01 — Product Scope](#) §7 for the full decision register with options + recommendations. Key decisions:

- **D1** — Primary obfuscating transport: MASQUE/QUIC (recommended).
- **D2** — Shared client-core language: Rust (make-or-break G3).
- **D3** — Event bus: Redis Streams MVP → NATS JetStream Phase 2.
- **D4** — Subnet collision: IPv6 ULA /48 + Tailscale 4via6.
- **D5** — Gateway edge language: Rust if G4 benchmark holds.
- **D6** — Transport topology: best-fit-per-leg as policy.
- **D7** — MVP ambition: lean tunnel-first.
- **D8** — Licensing / positioning: decide before public release.

Where to read next

- Product definition and scope → [01 — Product Scope](#)
- C4 architecture and components → [02 — System Architecture](#)
- Detailed roadmap and WBS → [11 — Guides & FAQs](#)
- Gap analysis → [docs/reviews/mvp-final/findings/phase1-docs-gap-analysis.md](#)

*Sources: docs/research/mvp/final/SPECIFICATION.md §1–§9, MASTER_IN
DEX.md, 00-product-scope-and-principles.md §1–§11.*